

# Supporting Information

## Characterizing Transport Properties of Surface Charged Nanofiltration Membranes via Model-based Data Analytics

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## Akaike Information Criterion (AIC) Derivations

### Log-Likelihood

Let  $\hat{y}_{r,i} = f(\mathbf{x}_{r,i}, \mathbf{u}_{r,i}, \boldsymbol{\theta})$  represent the model prediction for the  $i$ -th experimental observation  
1205  $y_{r,i}$ , where  $\mathbf{x}_{r,i}$  assembles state variables,  $\mathbf{u}_{r,i}$  are controlled experimental conditions,  $\boldsymbol{\theta}$  include  
all model parameters, and  $r \in R$  indexes the type of response. The model assumes that  
measurement errors are independent and normally distributed but may exhibit non-constant  
variance (heteroscedasticity) across different types of measurements. Specifically, the error,

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$\epsilon_{r,i}$ , associated with observation  $y_{r,i}$  is described as:

$$y_{r,i} = \underbrace{f(\mathbf{x}_{r,i}, \mathbf{u}_{r,i}, \boldsymbol{\theta})}_{\hat{y}_{r,i}} + \epsilon_{r,i}, \quad \epsilon_{r,i} \sim \mathcal{N}(0, s_{r,i}^2) \quad (\text{S1})$$

1210 where  $\mathcal{N}(0, s_{r,i}^2)$  represents a normal distribution with mean zero and variance  $s_{r,i}^2$ .

The probability distribution function of model observation  $y_{r,i}$  conditioned on parameter  $\boldsymbol{\theta}$  is given by

$$p(y_{r,i} | f(\mathbf{x}_{r,i}, \mathbf{u}_{r,i}, \boldsymbol{\theta}), s_{r,i}^2) = (2\pi s_{r,i}^2)^{-\frac{1}{2}} \exp\left(-\frac{(y_{r,i} - \hat{y}_{r,i})^2}{2s_{r,i}^2}\right) \quad (\text{S2})$$

The likelihood function, defined as a function of  $\boldsymbol{\theta}$  given a collection of observations  $\mathbf{y}$ , is the joint probability of these independent observations:

$$\mathcal{L}(\boldsymbol{\theta} | \mathbf{y}) = \prod_{r=1}^R \mathcal{L}(\boldsymbol{\theta} | \mathbf{y}_r) = \prod_{r=1}^R \prod_{i=1}^{n_r} p(y_{r,i} | f(\mathbf{x}_{r,i}, \mathbf{u}_{r,i}, \boldsymbol{\theta}), s_{r,i}^2) \quad (\text{S3})$$

1215 Taking the logarithm of the likelihood function, we have

$$\begin{aligned} \ln(\mathcal{L}(\boldsymbol{\theta} | \mathbf{y})) &= \sum_{r=1}^R \ln(\mathcal{L}(\boldsymbol{\theta} | \mathbf{y}_r)) \\ &= \sum_{r=1}^R \sum_{i=1}^{n_r} \ln p(y_{r,i} | f(\mathbf{x}_{r,i}, \mathbf{u}_{r,i}, \boldsymbol{\theta}), s_{r,i}^2) \\ &= -\frac{1}{2} \sum_{r=1}^R \sum_{i=1}^{n_r} \left( \ln(2\pi) + \ln s_{r,i}^2 + \frac{(y_{r,i} - \hat{y}_{r,i})^2}{s_{r,i}^2} \right) \\ &= -\frac{1}{2} \sum_{r=1}^R \left( n_r \ln(2\pi) + \sum_{i=1}^{n_r} \ln s_{r,i}^2 + \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{s_{r,i}^2} \right) \end{aligned} \quad (\text{S4})$$

## AIC with Constant Measurement Errors

When  $r$ -th type of measurements is subject to independent and identically distributed measurement error with constant variance, e.g.,  $\epsilon_r \sim \mathcal{N}(0, s_r^2)$ , the log-likelihood for  $r$ -th type

of measurements becomes:

$$\ln(\mathcal{L}(\boldsymbol{\theta} | \mathbf{y}_r)) = -\frac{1}{2} \left( n_r \ln(2\pi) + n_r \ln s_r^2 + \frac{1}{s_r^2} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) \quad (\text{S5})$$

1220 Equation S5 is derived by substituting  $s_{i,r} = s_r^2$  into eq S4.

At the maximum log-likelihood estimate (MLE), the necessary condition is:

$$\frac{\partial \ln(\mathcal{L}(\boldsymbol{\theta} | \mathbf{y}_r))}{\partial s_r} = -\frac{n_r}{s_r} + \frac{\sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2}{s_r^3} = 0 \quad (\text{S6})$$

which gives:

$$\hat{s}_r^2 = \frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \quad (\text{S7})$$

Substituting eq S7 into eq S5 gives:

$$\begin{aligned} \ln(\mathcal{L}(\hat{\boldsymbol{\theta}} | \mathbf{y}_r)) &= -\frac{1}{2} \left( n_r \ln(2\pi) + n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) + \frac{\sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2}{\frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2} \right) \\ &= -\frac{1}{2} \left( n_r \ln(2\pi) + n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) + n_r \right) \\ &= -\frac{1}{2} \left( n_r (\ln(2\pi) + 1) + n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) \right) \end{aligned} \quad (\text{S8})$$

In this case, AIC summed over all  $R$  types of measurements is

$$\begin{aligned} \text{AIC}^{\text{const}} &= -2 \ln(\mathcal{L}(\hat{\boldsymbol{\theta}} | \mathbf{y})) + 2k \\ &= (\ln(2\pi) + 1) \sum_{r=1}^R n_r + \sum_{r=1}^R \left( n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{r,i})^2 \right) \right) + 2k \end{aligned} \quad (\text{S9})$$

1225  $(\ln(2\pi) + 1) \sum_{r=1}^R n_r$  is a constant term across all models. Therefore, under constant errors

assumption, a practical formula for calculating AIC difference  $\Delta$  is:

$$\begin{aligned}\Delta_{a,b} &= \text{AIC}_a - \text{AIC}_b = \Delta\text{AIC}_a - \Delta\text{AIC}_b \\ \Delta\text{AIC}_a^{\text{const}} &= \sum_{r=1}^R \left( n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} (y_{r,i} - \hat{y}_{a,r,i})^2 \right) \right) + 2k_a\end{aligned}\quad (\text{S10})$$

## AIC with Proportional Measurement Errors

When  $r$ -th type of measurements are subject to a proportional (a.k.a., relative) error, e.g.,  $\epsilon_r \sim \mathcal{N}(0, \alpha_r^2 y_{r,i}^2)$ , the log-likelihood for measurement  $r$  becomes:

$$\ln(\mathcal{L}(\boldsymbol{\theta} \mid \mathbf{y}_r)) = -\frac{1}{2} \left( n_r \ln(2\pi) + n_r \ln \alpha_r^2 + \sum_{i=1}^{n_r} \ln y_{r,i}^2 + \frac{1}{\alpha_r^2} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2} \right) \quad (\text{S11})$$

1230 Equation S11 is derived by substituting  $s_{r,i}^2 = \alpha_r^2 y_{r,i}^2$  into eq S4.

At the maximum log-likelihood estimate, the necessary condition is:

$$\frac{\partial \ln(\mathcal{L}(\boldsymbol{\theta} \mid \mathbf{y}_r))}{\partial \alpha_r} = -\frac{n_r}{\alpha_r} + \frac{\sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2}}{\alpha_r^3} = 0 \quad (\text{S12})$$

which gives:

$$\hat{\alpha}_r^2 = \frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2} \quad (\text{S13})$$

Substituting eq S13 into eq S11 gives:

$$\begin{aligned}\ln(\mathcal{L}(\hat{\boldsymbol{\theta}} \mid \mathbf{y}_r)) &= -\frac{1}{2} \left( n_r \ln(2\pi) + n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2} \right) + \sum_{i=1}^{n_r} \ln y_{r,i}^2 + \frac{\sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2}}{\frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2}} \right) \\ &= -\frac{1}{2} \left( n_r \ln(2\pi) + n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2} \right) + \sum_{i=1}^{n_r} \ln y_{r,i}^2 + n_r \right) \\ &= -\frac{1}{2} \left( n_r (\ln(2\pi) + 1) + n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2} \right) + \sum_{i=1}^{n_r} \ln y_{r,i}^2 \right)\end{aligned}\quad (\text{S14})$$

In this case, AIC summed over all  $R$  types of measurements is

$$\begin{aligned} \text{AIC}^{\text{prop}} &= -2 \ln \left( \mathcal{L} \left( \hat{\boldsymbol{\theta}} \mid \mathbf{y} \right) \right) + 2k \\ &= (\ln(2\pi) + 1) \sum_{r=1}^R n_r + \sum_{r=1}^R \left( n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{r,i})^2}{y_{r,i}^2} \right) + \sum_{i=1}^{n_r} \ln y_{r,i}^2 \right) + 2k \end{aligned} \quad (\text{S15})$$

1235  $(\ln(2\pi) + 1) \sum_{r=1}^R n_r$  and  $\sum_{r=1}^R \sum_{i=1}^{n_r} \ln y_{r,i}^2$  (measurements) are constant terms across all models. Therefore, under proportional errors assumption, a practical formula for calculating AIC difference  $\Delta$  is:

$$\begin{aligned} \Delta_{a,b} &= \text{AIC}_a - \text{AIC}_b = \Delta \text{AIC}_a - \Delta \text{AIC}_b \\ \Delta \text{AIC}_a^{\text{prop}} &= \sum_{r=1}^R \left( n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} \frac{(y_{r,i} - \hat{y}_{a,r,i})^2}{y_{r,i}^2} \right) \right) + 2k_a \end{aligned} \quad (\text{S16})$$

## AIC Generalization

Next, we generalize the AIC formula for constant or proportional error assumptions. We  
1240 separate  $R$  into  $R_1$  and  $R_2$  accommodates the two types of measurement error structures.

$$\begin{aligned} \ln (\mathcal{L} (\boldsymbol{\theta} \mid \mathbf{y})) &= \sum_{r=1}^R \ln (\mathcal{L} (\boldsymbol{\theta} \mid \mathbf{y}_r)) \\ &= \sum_{r_1=1}^{R_1} \ln (\mathcal{L} (\boldsymbol{\theta} \mid \mathbf{y}_{r_1})) + \sum_{r_2=1}^{R_2} \ln (\mathcal{L} (\boldsymbol{\theta} \mid \mathbf{y}_{r_2})) \end{aligned} \quad (\text{S17})$$

Substituting eqs S8 and S14 into eq S17, we have

$$\begin{aligned}
\text{AIC} &= -2 \ln \left( \mathcal{L} \left( \hat{\boldsymbol{\theta}} \mid \mathbf{y} \right) \right) + 2k \\
&= \sum_{r_1=1}^{R_1} \left( n_{r_1} (\ln(2\pi) + 1) + n_{r_1} \ln \left( \frac{1}{n_{r_1}} \sum_{i=1}^{n_{r_1}} (y_{r_1,i} - \hat{y}_{r_1,i})^2 \right) \right) \\
&\quad + \sum_{r_2=1}^{R_2} \left( n_{r_2} (\ln(2\pi) + 1) + n_{r_2} \ln \left( \frac{1}{n_{r_2}} \sum_{i=1}^{n_{r_2}} \frac{(y_{r_2,i} - \hat{y}_{r_2,i})^2}{y_{r_2,i}^2} \right) + \sum_{i=1}^{n_{r_2}} \ln y_{r_2,i}^2 \right) + 2k \\
&= (\ln(2\pi) + 1) \sum_{r=1}^R n_r + \sum_{r_1=1}^{R_1} \left( n_{r_1} \ln \left( \frac{1}{n_{r_1}} \sum_{i=1}^{n_{r_1}} (y_{r_1,i} - \hat{y}_{r_1,i})^2 \right) \right) \\
&\quad + \sum_{r_2=1}^{R_2} \left( n_{r_2} \ln \left( \frac{1}{n_{r_2}} \sum_{i=1}^{n_{r_2}} \frac{(y_{r_2,i} - \hat{y}_{r_2,i})^2}{y_{r_2,i}^2} \right) + \sum_{i=1}^{n_{r_2}} \ln y_{r_2,i}^2 \right) + 2k
\end{aligned} \tag{S18}$$

Similar to eqs S10 and S16, by dropping constant terms, we have:

$$\begin{aligned}
\Delta \text{AIC} &= \sum_{r_1=1}^{R_1} \left( n_{r_1} \ln \left( \frac{1}{n_{r_1}} \sum_{i=1}^{n_{r_1}} (y_{r_1,i} - \hat{y}_{r_1,i})^2 \right) \right) \\
&\quad + \sum_{r_2=1}^{R_2} \left( n_{r_2} \ln \left( \frac{1}{n_{r_2}} \sum_{i=1}^{n_{r_2}} \frac{(y_{r_2,i} - \hat{y}_{r_2,i})^2}{y_{r_2,i}^2} \right) \right) + 2k \\
&= \sum_{r_1=1}^{R_1} \left( n_{r_1} \ln \left( \frac{\tilde{s}_{r_1}^2}{n_{r_1}} \sum_{i=1}^{n_{r_1}} \frac{(y_{r_1,i} - \hat{y}_{r_1,i})^2}{\tilde{s}_{r_1}^2} \right) \right) \\
&\quad + \sum_{r_2=1}^{R_2} \left( n_{r_2} \ln \left( \frac{\tilde{\alpha}_{r_2}^2}{n_{r_2}} \sum_{i=1}^{n_{r_2}} \frac{(y_{r_2,i} - \hat{y}_{r_2,i})^2}{\tilde{\alpha}_{r_2}^2 y_{r_2,i}^2} \right) \right) + 2k
\end{aligned} \tag{S19}$$

$\tilde{s}_{r_1}$  and  $\tilde{\alpha}_{r_2}$  represents initial guess for  $s_{r_1}$  and  $\alpha_{r_1}$  associated with the each type of measurements.

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We introduce scaled residuals  $z_{r,i}$  are computed by dividing the residuals by their corresponding standard deviations that we assumed,  $\tilde{s}$ :

$$z_{r,i} = \frac{y_{r,t} - \hat{y}_{r,t}}{\tilde{s}_{r,i}} \tag{S20}$$

where  $\tilde{s}_{r,i} = \tilde{s}_r$  for constant errors and  $\tilde{s}_{r,i} = \tilde{\alpha}_r y_{r,i}$  for proportional errors. Then eq S19 can

be expressed in  $z$ :

$$\begin{aligned}
\Delta\text{AIC} &= \sum_{r_1=1}^{R_1} \left( n_{r_1} \ln \left( \frac{\tilde{s}_{r_1}^2}{n_{r_1}} \sum_{i=1}^{n_{r_1}} z_{r_1,i}^2 \right) \right) + \sum_{r_2=1}^{R_2} \left( n_{r_2} \ln \left( \frac{\tilde{\alpha}_{r_2}^2}{n_{r_2}} \sum_{i=1}^{n_{r_2}} z_{r_2,i}^2 \right) \right) + 2k \\
&= \sum_{r_1=1}^{R_1} \left( n_{r_1} \ln \left( \frac{1}{n_{r_1}} \sum_{i=1}^{n_{r_1}} z_{r_1,i}^2 \right) + n_{r_1} \ln \tilde{s}_{r_1}^2 \right) \\
&\quad + \sum_{r_2=1}^{R_2} \left( n_{r_2} \ln \left( \frac{1}{n_{r_2}} \sum_{i=1}^{n_{r_2}} z_{r_2,i}^2 \right) + n_{r_2} \ln \tilde{\alpha}_{r_2}^2 \right) + 2k \\
&= \sum_{r=1}^R \left( n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} z_{r,i}^2 \right) \right) + \sum_{r_1=1}^{R_1} (n_{r_1} \ln \tilde{s}_{r_1}^2) + \sum_{r_2=1}^{R_2} (n_{r_2} \ln \tilde{\alpha}_{r_2}^2) + 2k
\end{aligned} \tag{S21}$$

$\tilde{s}_{r_1}$  and  $\tilde{\alpha}_{r_2}$  are predetermined and same across all models, therefore, a generalized  $\Delta\text{AIC}$  robust to a mixture of the two types of errors can be calculated through:

$$\Delta\text{AIC} = \sum_{r=1}^R \left( n_r \ln \left( \frac{1}{n_r} \sum_{i=1}^{n_r} z_{r,i}^2 \right) \right) + 2k \tag{S22}$$

We caution this formula may not hold if the weights  $\tilde{s}_{r_1}$  and  $\tilde{\alpha}_{r_2}$  are updated by iteratively alternating between WLS and then setting  $\tilde{s}_{r_1} \leftarrow \hat{s}_{r_1}$  and  $\tilde{\alpha}_{r_2} \leftarrow \hat{\alpha}_{r_2}$ . We did not perform iterative WLS in this work. With iterative WLS, the assumption that  $\tilde{s}_{r_1}$  and  $\tilde{\alpha}_{r_2}$  are the same across all models may not be valid.

## Relate Partition Coefficient with Donnan Equilibria

### Zeta Potential Measurements

Three trials using 2 cm  $\times$  1 cm squares of membrane were conducted. The samples were first soaked in deionized water overnight. The samples were placed parallel to one another, leaving a small gap between them, creating a rectangular channel for the electrolyte solution to pass through. A 1 mM KCl solution at pH 2 is used to flush the space between the samples. Afterwards, a select volume of 1 mM KCl at pH 2 is passed through the rectangular channel.

The pressure is ramped from 600 mbar to 200 mbar and the voltage along the pressure ramp is recorded. Using voltage, pressure, conductivity, and viscosity of the solution, a zeta potential is calculated. This sequence is repeated three times at pH values ranging from 2 to 11.

## Surface Charge Density and Partition Coefficient Analysis

Figure S9 depicts the zeta potential measurements for an NF270 membrane. Diafiltration experiments were conducted at the unadjusted pH of the KCl solutions ( $\sim$ pH 6). Taking the zeta potential measurement at pH 6, the Gouy-Chapman equation<sup>86</sup> can be used to calculate the surface charge density. Assuming the charge is uniformly distributed within the void volume of the pores, an effective volumetric charge density ( $\chi$ ) of 140 mM was calculated for the NF270 membranes.

Ion partitioning into nanofiltration membranes is impacted by a number of energetic considerations. Here, for the NF270 membranes, it is assumed that steric hindrance and electrostatic interactions are the dominant factors determining partitioning. Therefore, assuming equilibrium at the membrane-solution interface, and equating the electrochemical potential for each ion  $i$  inside and outside of the membrane:

$$RT \ln(c_i^m) + z_i F \Delta\phi_D + \Delta\mu_i = RT \ln(c_i^s) \quad (\text{S23})$$

where  $c^m$  is the solution-membrane interface concentration in the membrane phase,  $c^s$  is the interface concentration in the solution phase,  $z$  is the valency of the ion,  $F$  is the Faraday constant,  $\Delta\phi_D$  is the electrostatic potential difference and  $\Delta\mu$  is the chemical potential difference of the ion. This can be rearranged as

$$\frac{c_i^m}{c_i^s} = H_i \exp\left(\frac{-z_i F \Delta\mu_i}{RT}\right) \quad (\text{S24})$$

where  $H_i$  is the individual ion partition coefficient determined by the extent of steric hin-



drance. This expression can be written for both the co-ion,  $i$ , and the counter-ion,  $j$ , yielding the following constraint on the ion concentrations inside and outside of the membrane phase.

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$$\frac{c_i^m c_j^m}{c_i^s c_j^s} = H_i H_j = H_{\text{steric}} \quad (\text{S25})$$

Assuming a 1:1 salt, eq S25 can be combined with electroneutrality constraints in the membrane and solution phases to develop an expression for the co-ion concentration within the membrane phase.

$$c_i^m = \frac{\sqrt{\chi^2 + 4H_{\text{steric}}c_{\text{salt}}^2} - \chi}{2} \quad (\text{S26})$$

For small bulk concentrations ( $c_{\text{salt}} \ll \chi$ ), a Taylor series expansion is used to approxi-

1290 mate  $c_i^m$ :

$$c_i^m \approx \frac{H_{\text{steric}}c_{\text{salt}}^2}{\chi} \quad (\text{S27})$$

Dividing  $c_i^m$  by the bulk concentration of salt to calculate the partition coefficient equation, we obtain:

$$H = \frac{c_i^m}{c_{\text{salt}}} \approx \frac{H_{\text{steric}}}{\chi} c_{\text{salt}} \quad (\text{S28})$$

Our estimate  $H = h_0 + h_1 c_{\text{in},f}$  is consistent with the first-order dependence of  $H$  on the interface concentration in the solution phase. Assuming that  $H_{\text{steric}} = h_0$ ,  $h_1$  can be related to the partition coefficient described in equation eq S28 by substituting the membrane charge density ( $\chi = 140 \text{ mM}$ ).

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$$h_1 \approx \frac{1}{\chi} \approx \frac{h_0}{140} \approx 0.0042 \text{ mM}^{-1} \quad (\text{S29})$$

$h_1$  fitted from the diafiltration experiment is  $0.0011 \text{ mM}^{-1}$ , which is on the same order of magnitude as this calculated value. This agreement supports a hypothesis that a combination of steric hindrance and electrostatic interactions underlie the transport properties of NF270 membranes.

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## Supporting Tables & Figures

Table S1: Estimated model parameters for different variations of the solute transport equation used in Table 3 and 4. Values correspond to the best-fit parameters obtained from the lag experiment in Figure S2.

| B Form                                      | Parameters                                                                        |                                          |                                        |                                             |                               |                                             |
|---------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------|----------------------------------------|---------------------------------------------|-------------------------------|---------------------------------------------|
|                                             | Units                                                                             |                                          |                                        |                                             |                               |                                             |
|                                             | $L_p$<br>$\text{L} \cdot \text{m}^{-2} \cdot \text{h}^{-1} \cdot \text{bar}^{-1}$ | $B$<br>$\mu\text{m} \cdot \text{s}^{-1}$ | $\sigma$<br>dimensionless              | $\dot{V}$<br>$\text{g} \cdot \text{h}^{-1}$ |                               |                                             |
| $B = \text{const.}$                         | 10.11                                                                             | 16.42                                    | 1.00                                   | 0.10                                        |                               |                                             |
| $B_i = \text{const.}$                       |                                                                                   | 6.50                                     |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 18.47                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 14.64                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 13.82                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 13.47                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 12.71                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 16.61                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 18.97                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 19.93                                    |                                        |                                             |                               |                                             |
|                                             |                                                                                   | 21.82                                    |                                        |                                             |                               |                                             |
| $B = J_w \sum_{i \in I} \beta_i c_{in,i}^i$ | $L_p$<br>$\text{L} \cdot \text{m}^{-2} \cdot \text{h}^{-1} \cdot \text{bar}^{-1}$ | $\beta_0$<br>dimensionless               | $\beta_1$<br>$\text{mM}^{0.5/-0.5/-1}$ | $\beta_2$<br>$\text{mM}^{-2}$               | $\beta_3$<br>$\text{mM}^{-3}$ | $\sigma$<br>dimensionless                   |
|                                             | $I = \{0\}$                                                                       | 2.20                                     | -                                      | -                                           | -                             | 1.00                                        |
|                                             | $I = \{0, -0.5\}$                                                                 | 3.30                                     | -9.9137                                | -                                           | -                             | 1.00                                        |
|                                             | $I = \{0, 0.5\}$                                                                  | 0.30                                     | 0.2033                                 | -                                           | -                             | 1.00                                        |
|                                             | $I = \{0, 1\}$                                                                    | 1.00                                     | 0.0135                                 | -                                           | -                             | 1.00                                        |
|                                             | $I = \{0, 1, 2\}$                                                                 | 1.85                                     | -0.0187                                | $2.41 \times 10^{-4}$                       | -                             | 1.00                                        |
|                                             | $I = \{0, 1, 2, 3\}$                                                              | -1.42                                    | 0.1777                                 | $-3.08 \times 10^{-3}$                      | $1.69 \times 10^{-5}$         | 1.00                                        |
|                                             |                                                                                   |                                          |                                        |                                             |                               | $\dot{V}$<br>$\text{g} \cdot \text{h}^{-1}$ |

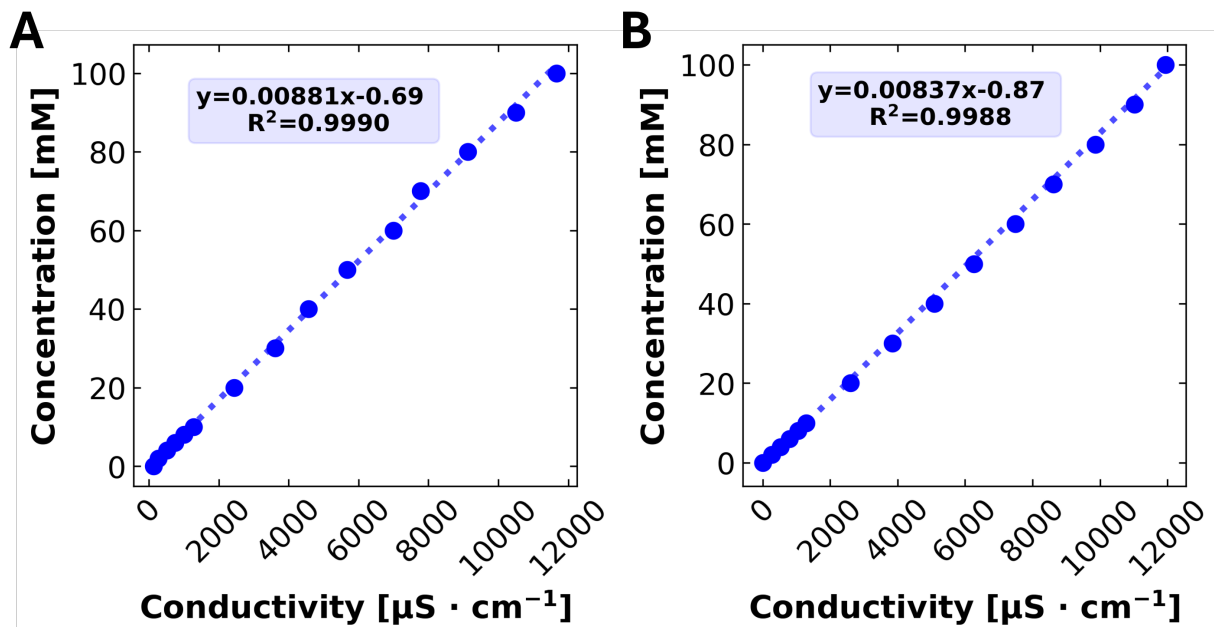


Figure S1: The calibration curves used to convert conductivity measurements into the corresponding KCl concentration for the (A) CN0359 conductivity board and (B) for EC EvalBoard LFS conductivity board. To enable the CN0359 conductivity board to be compatible with the LSF 1107 conductivity probe, the board needed to be modified (i.e., previously reported by Muetzel et al.<sup>54</sup>). To simplify the design, the EC EvalBoard LSF conductivity board, which is manufactured by the same company as the probes, is used. The conductivity was calculated using,  $C = GAV^{-1}$ , the relationship between the conductivity probe cell constant ( $G = 0.42 \text{ cm}^{-1}$ ), the peak to peak applied current, ( $A = 1 \text{ mA}$ ), and the peak to peak differential voltage drop,  $V$ .

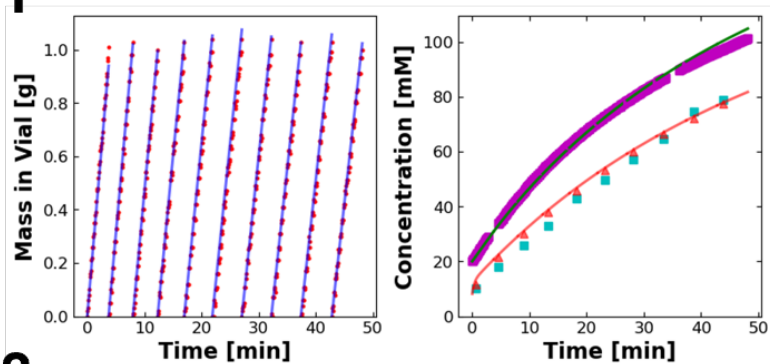
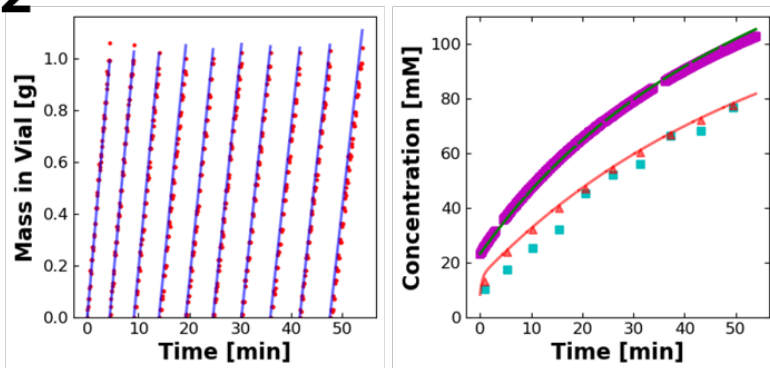
**A.1****A.2**

Figure S2: Empirical model prediction for additional experiments. (A.1) Diafiltration experiments were performed in the lag mode, at 60 psi, with 15 mM/120 mM KCl feed/diafiltrate solutions for the NF270 membrane second sample. (A.2) Diafiltration experiments were performed in the lag mode, at 60 psi, with 15 mM/120 mM KCl feed/diafiltrate solutions for the NF270 membrane third sample.

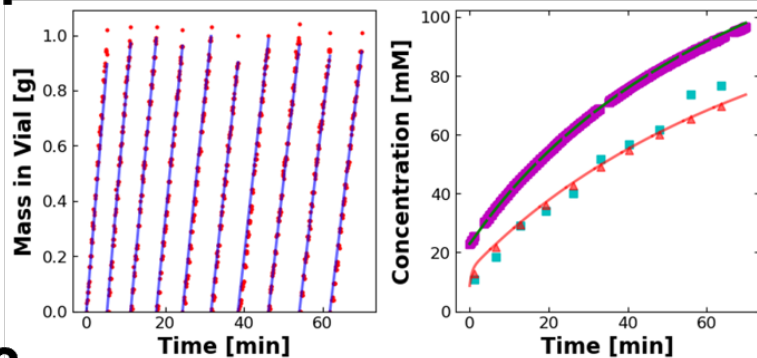
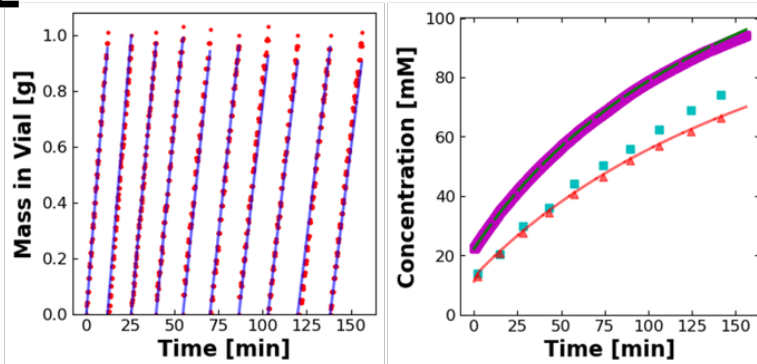
**B.1****B.2**

Figure S3: Empirical model prediction for additional experiments. (B.1) Diafiltration experiments were performed in the lag mode, at 40 psi, with 15 mM/120 mM KCl feed/diafiltrate solutions for the NF270 membrane sample. (B.2) Diafiltration experiments were performed in the lag mode, at 20 psi, with 15 mM/120 mM KCl feed/diafiltrate solutions for the NF270 membrane sample.

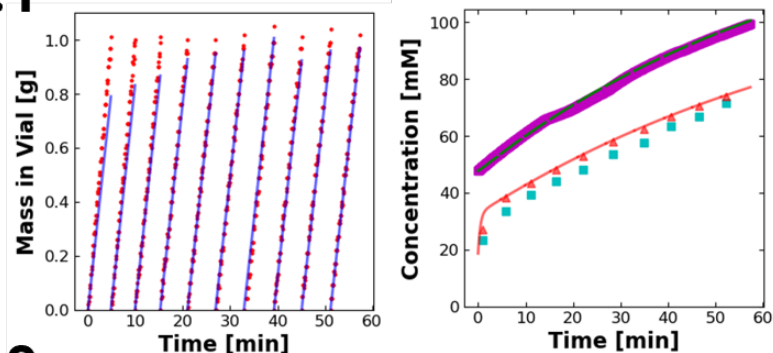
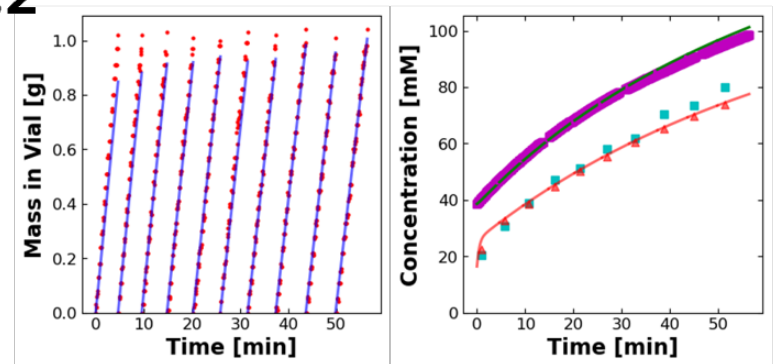
**C.1****C.2**

Figure S4: Empirical model prediction for additional experiments. (C.1) Diafiltration experiments were performed in the overflow mode, at 60 psi, with 15 mM/120 mM KCl feed/diafiltrate solutions for the NF270 membrane sample. (C.2) Diafiltration experiments were performed in the overflow mode, at 60 psi, with 15 mM/120 mM KCl feed/diafiltrate solutions for the NF270 membrane sample.

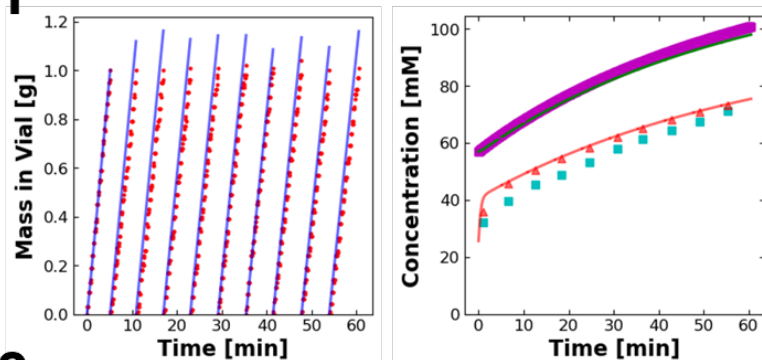
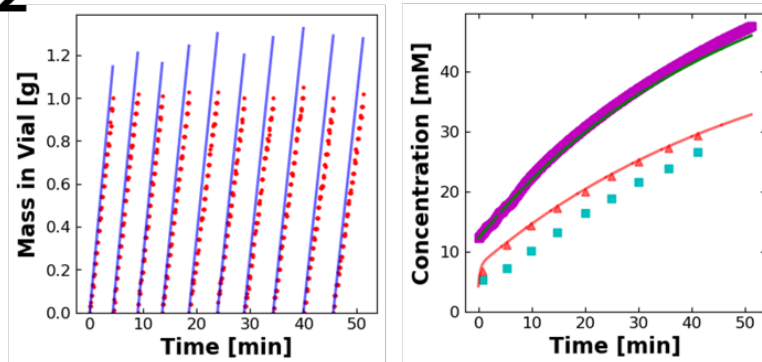
**D.1****D.2**

Figure S5: Empirical model prediction for additional experiments. (D.1) Diafiltration experiments were performed in the lag mode, at 60 psi, with 50 mM/100 mM KCl feed/diafiltrate solutions for the NF270 membrane sample. (D.2) Diafiltration experiments were performed in the lag mode, at 60 psi, with 10 mM/50 mM KCl feed/diafiltrate solutions for the NF270 membrane sample.



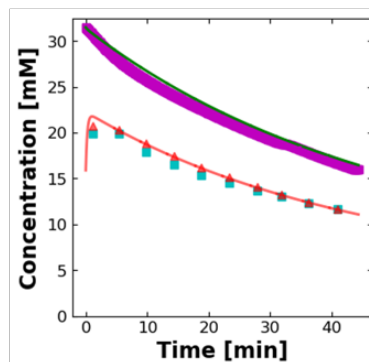
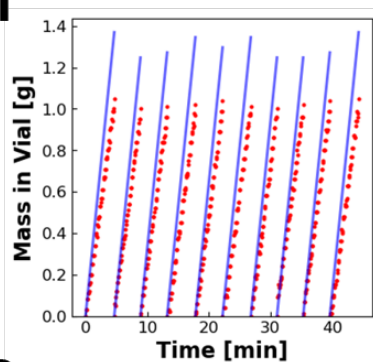
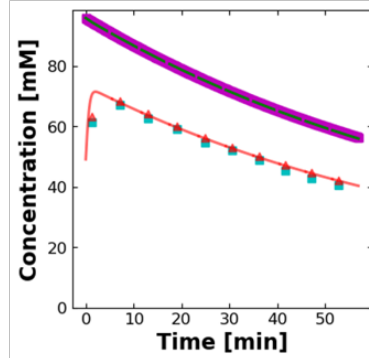
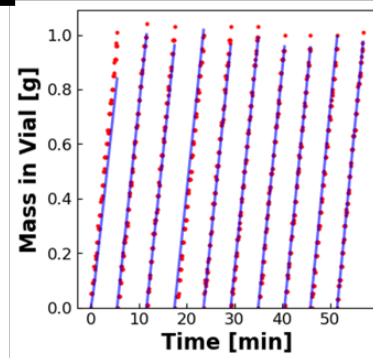
**E.1****E.2**

Figure S6: Empirical model prediction for additional experiments. (E.1) Diafiltration experiments were performed in the lag mode, at 60 psi, with 30 mM/1 mM KCl feed/diafiltrate solutions for the NF270 membrane sample. (E.2) Diafiltration experiments were performed in the lag mode, at 60 psi, with 100 mM/12 mM KCl feed/diafiltrate solutions for the NF270 membrane sample.

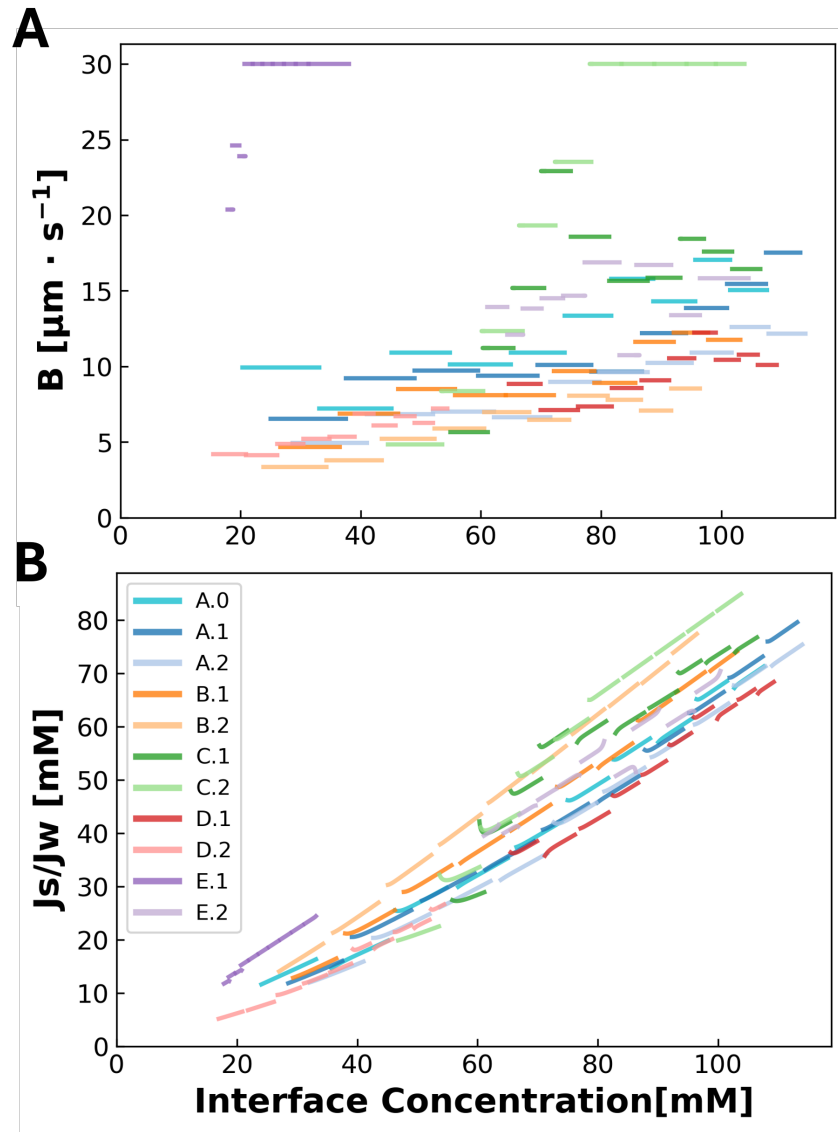


Figure S7: (A) Discrete  $B$  values fitted for each vial collection in diafiltration experiments, plotted against the feed-side solution-membrane interface concentration. (B) Solute flux normalized by water flux ( $J_s/J_w$ ) plotted against interface concentration. Diafiltration experiment conditions are provided in Figure 6A and B for A.0, and in Figure S2-S6 for the remaining cases.

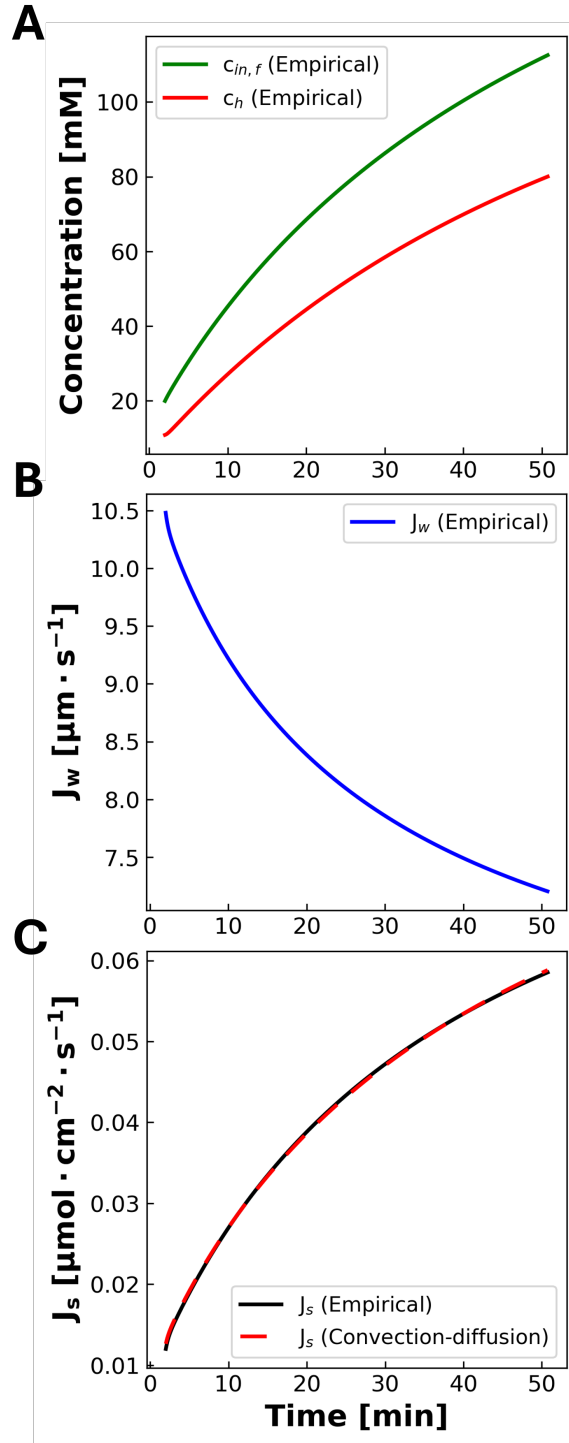


Figure S8: Plots illustrating the fitting of the convection-diffusion model (eq 25) to simulated data from the empirical model (eq 31). Solid lines in panels (A) ( $c_{in,f}$  and  $c_h$ ), (B) ( $J_w$ ), and (C) ( $J_s$ ) represent simulated data from the empirical model for a lag diafiltration experiment. The red dashed line in (C) depicts the solute flux  $J_s$  obtained from the convection-diffusion model fit. Corresponding model parameters are reported in Table 2 and Figure 8.

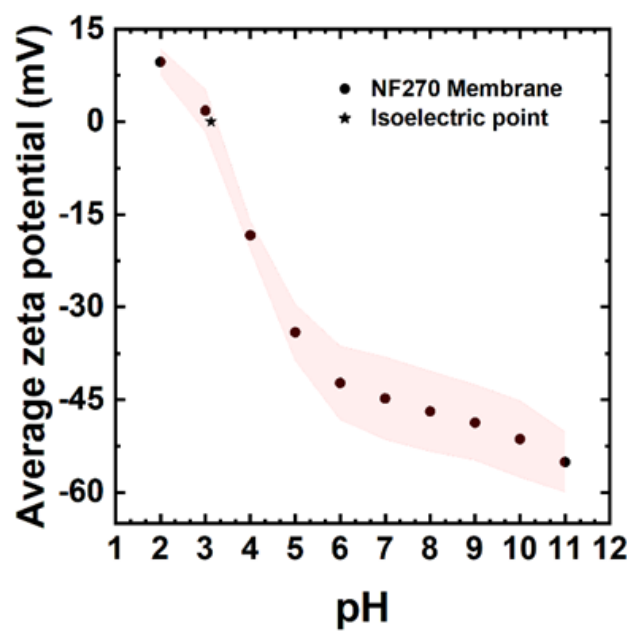


Figure S9: Zeta potential measured for NF270.